

Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030414

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Query 1: Create the database

```
2 CREATE DATABASE IF NOT EXISTS bookflow5_db;
```

Output: -

✓	2	10:09:03	CREATE DATABASE IF NOT EXISTS bookflow5_db	1 row(s) affected	0.094 sec
---	---	----------	--	-------------------	-----------

Query 2: Select the database

```
USE bookflow5_db;
```

Output: -

✓	3	10:11:29	USE bookflow5_db	0 row(s) affected	0.000 sec
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Query 3: Create the books table

```
> CREATE TABLE books (  
  book_id INT AUTO_INCREMENT PRIMARY KEY,  
  title VARCHAR(255) NOT NULL,  
  isbn VARCHAR(13) NOT NULL UNIQUE,  
  published_year INT,  
  CONSTRAINT chk_published_year CHECK (published_year<2027)  
);
```

Output: -

✓	4	10:18:46	CREATE TABLE books (book_id INT AUTO_INCR...	0 row(s) affected	0.125 sec
---	---	----------	--	-------------------	-----------

Query 4: Create the members table

```
⊖ CREATE TABLE members (  
  member_id INT AUTO_INCREMENT PRIMARY KEY,  
  full_name VARCHAR(100) NOT NULL,  
  email VARCHAR(150) NOT NULL UNIQUE  
);
```

Output: -

✓	5	10:23:27	CREATE TABLE members (member_id INT AUTO_I...	0 row(s) affected	0.079 sec
---	---	----------	--	-------------------	-----------

Query 5 (verification): Inspect the table structures

```
);  
DESCRIBE books;
```

Output: -

✓	6	10:26:25	DESCRIBE books	4 row(s) returned	0.047 sec / 0.000 sec
---	---	----------	----------------	-------------------	-----------------------

Result Grid Filter Rows: Export: Wrap Cell Content:						
	Field	Type	Null	Key	Default	Extra
▶	book_id	int	NO	PRI	NULL	auto_increment
	title	varchar(255)	NO		NULL	
	isbn	varchar(13)	NO	UNI	NULL	
	published_year	int	YES		NULL	

Query 6: Insert 3 books

```
INSERT INTO books (title,isbn,published_year) VALUES  
( 'The Alchemist', '9780061122415', 1988),  
( 'The autobigraphy', '9780132350884', 2005),  
( 'Harry Potter', '9780735211292', 1997);
```

Output: -

✓	8	10:33:11	INSERT INTO books (title,isbn,published_year) VAL...	3 row(s) affected Records: 3 Duplicates: 0 Warning...	0.031 sec
---	---	----------	--	---	-----------

Query 7: Insert 3 members

```
INSERT INTO members(full_name, email) VALUES  
( 'Yishaani', 'yishaani@gmail.com'),  
( 'Rahul Sharma', 'rahul@gmail.com'),  
( 'Priya Reddy', 'priya@gmail.com');
```

Output: -

✓	9	10:38:04	INSERT INTO members (full_name,email) VALUES (...	3 row(s) affected Records: 3 Duplicates: 0 Warning...	0.047 sec
---	---	----------	---	---	-----------

Query 8: Display

```
SELECT * FROM books;
```

Output: -

```
10 10:40:26 SELECT * FROM books LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
```

Result Grid

Filter Rows:

Edit:

	book_id	title	isbn	published_year
▶	1	The Alchemist	9780061122415	1988
	2	The autobiography	9780132350884	2005
	3	Harry Potter	9780735211292	1997
✱	NULL	NULL	NULL	NULL

```
SELECT * FROM members;
```

```
11 10:41:15 SELECT * FROM members LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
```

2	Adhitha	adhitha@gmail.com
3	Nihitha	nihitha@gmail.com
NULL	NULL	NULL

Query 9: Constraint Tests (These MUST Fail)

Test 1: Duplicate ISBN (UNIQUE violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES ('Fake Copy', '9780061122415', 2000);
```

Output: -

```
12 10:47:42 INSERT INTO books (title, isbn, published_year) VALUES... Error Code: 1062. Duplicate entry '9780061122415' for key... 0.047 sec
```

Test 2: NULL title (NOT NULL violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES (NULL, '9999999999999', 2010);
```

Output: -

```
14 10:50:07 INSERT INTO books (title, isbn, published_year) VALUES (NULL, '9999999999999', 2010) Error Code: 1048. Column 'title' cannot be null 0.000 sec
```

Test 3: Future publication year (CHECK violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES ('Time Traveler', '8888888888888', 2030);
```

Output: -

Test 4: Duplicate email (UNIQUE violation)

```
INSERT INTO members (full_name, email)
VALUES ('Adhritha Reddy', 'adhritha@gmail.com');
```

Output: -